Edge-connectivity thresholds forcing positive Lin-Lu-Yau curvature

Ziming Zhou, Shenggui Zhang

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Editorial note. Complete original mathematical body125–1107, all proof/case work, original curvature figure and embedded bibliography retained. Authors retain responsibility for mathematics; no proof repair, extra hypothesis, independent counterexample search, author review or certification.

Editorial note. Source0 changes only four non-mathematical preamble comments containing Chinese text to ASCII comments, retaining the original percent markers and all native commands/line positions. Original main.tex retained unchanged; no font installation or mathematical adaptation introduced.

Editorial note. Curvature scale2/3 cases form one object; order/size threshold formulations and original sharpness statements/proofs form one other object. Direct consequences and examples are not separately counted. Original statement quantifiers and native labels remain as written.

Editorial note. Actual missing floatrow dependency supplied locally from original official TeX Live distribution/source/license; raw source and failed r001 preserved, no global installation.

Editorial note. Actual t1xr missing curly double quotation glyphs rendered as equivalent TeX opening/closing quotes in source0; raw main.tex unchanged and all line positions preserved.

Editorial note. Remaining actual t1xr missing apostrophe/en-dash glyphs at raw259 and bibliography1088 rendered with equivalent TeX apostrophe/dash in source0, preserving line positions and raw source.

Allowed standard prerequisites (editorial).

- Edge cuts, optimal transportation and scale-dependent Lin-Lu-Yau curvature. Original connectedness/diameter/minimum degree/order/size hypotheses, Wasserstein and idleness conventions, Kantorovich formulas and all source-local auxiliary lemmas retained with original standard citations.. Both curvature-to-connectivity and connectivity-to-curvature directions are retained with full original proof/case work and original extremal constructions, without independently verifying or repairing author statements.

1 Introduction

A graph is *connected* if, for every partition of its vertex set into two nonempty sets X and Y, there is an edge with one end in X and one end in Y; otherwise it is *disconnected*. Connectivity is one of the fundamental concepts of graph theory. There are two parameters of a graph, its connectivity and edge-connectivity, which measure the extent to which it is connected [1, 10].

The *connectivity* $\kappa(G)$ of a non-complete graph G is defined as the minimum number of vertices whose removal disconnects the graph. The *edge-connectivity* $\kappa'(G)$ of a graph G with at least two vertices is defined as the minimum number of edges whose removal disconnects the graph. By convention, the connectivity of a complete graph K_n with n vertices is defined to be $n-1$, and the edge-connectivity of a graph with only a single vertex is defined to be 0.

A classical bound on connectivity due to Whitney [22] reads as

$$\kappa(G) \le \kappa'(G) \le \delta(G),$$

where $\delta(G)$ denotes the minimum vertex degree of G. Both inequalities are sharp. For instance, the identity $\kappa(G) = \kappa'(G) = \delta(G)$ holds for any finite connected edge-transitive graph [18, 21], for any finite distance-regular graph [3–5], and for any finite $(0, 2)$-graph [6]. In general, these three quantities $\kappa(G)$, $\kappa'(G)$, and $\delta(G)$ can differ by arbitrarily large amounts. Indeed, for any integers $0 < \kappa \leq \kappa' \leq \delta$, there exists a graph G with $\kappa(G) = \kappa$, $\kappa'(G) = \kappa'$, and $\delta(G) = \delta$.

For over a century, Ricci curvature has been a fundamental concept in the study of smooth Riemannian manifolds, governing volume growth and geodesic behavior. Because of its importance, there has been considerable interest in generalizing Ricci curvature to discrete structures in recent decades. In 2009, Ollivier [19] introduced a notion of coarse Ricci curvature for Markov chains on metric spaces via optimal transport. This notion is particularly well suited for locally finite graphs. Lin, Lu, and Yau [15] modified Ollivier's definition to introduce the Lin–Lu–Yau curvature on graphs in 2011. This refinement is obtained by differentiating the Ollivier-Ricci curvature with respect to the idleness parameter. The Lin–Lu–Yau curvature detects local structural information of a graph and has been widely studied in recent years.

The Lin–Lu–Yau curvature is defined for any two distinct vertices x, y of a graph G. When x and y are adjacent, we call $\kappa_{\mathrm{LLY}}(x, y)$ the Lin–Lu–Yau curvature of the edge xy. The precise definition is formulated via the Wasserstein distance between probability measures; see Subsection 2.2. In Figure 1, the number on each edge is the Lin–Lu–Yau curvature of that edge, computed by the graph curvature calculator [8].

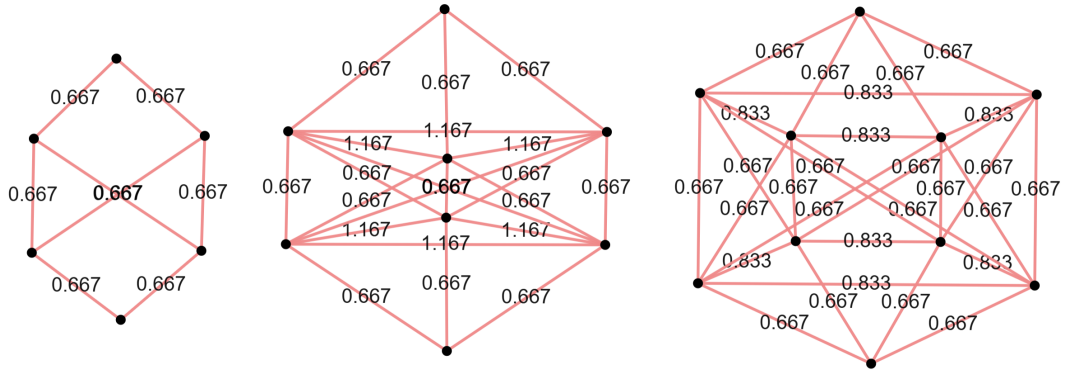


Fig. 1: Three graphs labeled with the Lin–Lu–Yau curvature of each edge. The figure is produced by the interactive app "Graph curvature calculator" at https://www.mas.ncl.ac.uk/graph-curvature/.

For two vertices x, y in G, we denote by $d(x, y)$ the distance between x and y. For any positive integer i, let us denote the lower Lin–Lu–Yau curvature bound at scale i by

$$\kappa_{\mathrm{LLY}}^{(i)}(G) := \inf_{\substack{x,y \in V \\ d(x,y)=i}} \kappa_{\mathrm{LLY}}(x, y).$$

When $i = 1$, we simply write $\kappa_{\mathrm{LLY}}(G)$ instead of $\kappa_{\mathrm{LLY}}^{(1)}(G)$.

Chen, Liu and You [7] showed that positive Lin–Lu–Yau curvature implies large connectivity. They proved that $\kappa(G) \geq \delta(G)\kappa_{\mathrm{LLY}}^{(2)}(G)$ holds for every non-complete connected graph G with minimum degree $\delta(G)$ and connectivity $\kappa(G)$. They further characterized the structure of an infinite graph family for which equality is attained in the inequality.

Our first two results show that positive Lin–Lu–Yau curvature implies large edge-connectivity.

Theorem 1.1. *Let G be a connected graph with diameter* $\mathrm{diam}(G) = 2$*, minimum degree $\delta(G)$ and edge-connectivity $\kappa'(G)$. Then,*

$$\kappa'(G) \geq \delta(G)\kappa_{\mathrm{LLY}}^{(2)}(G).$$

Theorem 1.2. *Let G be a connected graph with diameter* $\mathrm{diam}(G) \geq 3$*, minimum degree $\delta(G)$ and edge-connectivity $\kappa'(G)$. Then,*

$$\kappa'(G) \geq \frac{3}{2}\delta(G)\kappa_{\mathrm{LLY}}^{(3)}(G).$$

Recall that $\kappa^{(3)}_{\mathrm{LLY}}(G) \geq \kappa^{(2)}_{\mathrm{LLY}}(G) \geq \kappa_{\mathrm{LLY}}(G)$ by the triangle inequality of the Wasserstein distance; see [15, Lemma 2.3] or Lemma 2.2 below. Therefore, Theorem 1.1 implies in particular that

$$\kappa'(G) \geq \delta(G)\kappa_{\mathrm{LLY}}(G)$$

for a graph with diameter $\mathrm{diam}(G) = 2$, while Theorem 1.2 implies that

$$\kappa'(G) \geq \frac{3}{2}\delta(G)\kappa_{\mathrm{LLY}}(G)$$

for a graph with diameter $\mathrm{diam}(G) \geq 3$.

For any graph G obtained by deleting a non-empty matching from a complete graph K_n with $n \geq 3$ vertices, the equality $\kappa'(G) = \delta(G)\kappa_{\mathrm{LLY}}(G)$ holds, and hence equality also holds in Theorem 1.1. Indeed, it was proved independently in [9, Theorem 1.3] and [14, Theorem 1] that $\kappa_{\mathrm{LLY}}(G) = 1$ for such a graph. Therefore, $\kappa'(G) \geq \delta(G)$, which forces equality.

For the three graphs in Figure 1, the equality $\kappa'(G) = \frac{3}{2}\delta(G)\kappa_{\mathrm{LLY}}(G)$ holds, and hence equality also holds in Theorem 1.2. Moreover, let $G_{2n} \cong K_n \cup K_n + M_2$ be the union of two copies of the complete graph K_n ($n \geq 3$) with a matching of size 2 between them. Equality holds in Theorem 1.2 for G_{2n}; see Lemma 3.3 below. In fact, the graphs G_{2n} form an infinite family for which Theorem 1.2 is sharp.

In Riemannian geometry, spaces with positive Ricci curvature enjoy numerous properties, many of which have natural probabilistic interpretations. We refer to the comprehensive survey [17] and the references therein for a discussion of the geometric interest of lower bounds on Ricci curvature and the need for generalized notions of positive Ricci curvature for metric measure spaces.

Hehl [12] established a lower bound on the minimum degree that guarantees non-negative Lin–Lu–Yau curvature. He proved that if $\delta(G) \geq \frac{2n}{3} - 1$, then $\kappa_{\mathrm{LLY}}(G) \geq 0$. Moreover, the bound $\frac{2n}{3} - 1$ is optimal. There is also a lower bound on the connectivity that guarantees positive Lin–Lu–Yau curvature: namely, if $\kappa(G) \geq \frac{n-1}{2}$, then $\kappa_{\mathrm{LLY}}(G) > 0$, and the bound $\frac{n-1}{2}$ is optimal; see [7, Theorem 1.7 and Theorem 1.8].

We derive sufficient conditions on edge-connectivity for positive Lin–Lu–Yau curvature. The following results show that sufficiently large edge-connectivity implies large Lin–Lu–Yau curvature.

Theorem 1.3. *Let G be a connected graph with n vertices and edge-connectivity $\kappa'(G)$. If*

$$\kappa'(G) \geq \frac{2n}{3} - 1,$$

then $\kappa_{\mathrm{LLY}}(x, y) > 0$ for every edge xy of G.

After deriving this lower bound on the edge-connectivity of a graph that guarantees positive Lin–Lu–Yau curvature, we proceed to discuss the sharpness of the bound. We notice that the lower bound $\frac{2n}{3} - 1$ is optimal. There exist infinitely many graphs with edge-connectivity $\frac{2n}{3} - 2$, each containing an edge of negative Lin–Lu–Yau curvature. Precisely, we have the following result.

Theorem 1.4. *Let $n \in \mathbb{N}$ such that $2n/3 \in 2\mathbb{N}$ and $2n/3 \geq 6$. Then there exist infinitely many graphs with n vertices and edge-connectivity*

$$\kappa'(G) = \frac{2n}{3} - 2$$

that contain an edge xy with $\kappa_{\mathrm{LLY}}(x, y) < 0$.

We derive a sufficient condition, depending only on the number of edges, for positive Lin–Lu–Yau curvature.

Theorem 1.5. *Let G be a connected graph with m edges and edge-connectivity $\kappa'(G)$. If*

$$\kappa'(G) \geq \left\lfloor \frac{\sqrt{1 + \frac{16}{3}m} - 1}{2} \right\rfloor,$$

then $\kappa_{\mathrm{LLY}}(x, y) > 0$ for every edge xy of G.

We also discuss the sharpness of the bound.

Theorem 1.6. *Let $m \in \mathbb{N}$ such that $m \geq 11$. Then there exist infinitely many graphs with m edges and edge-connectivity*

$$\kappa'(G) = \left\lceil \frac{\sqrt{1 + \frac{16}{3}m} - 1}{2} \right\rceil - 1$$

that contain an edge xy with $\kappa_{\mathrm{LLY}}(x, y) = 0$.

2 Preliminaries

In this section, we revisit key concepts from Graph Theory and Optimal Transport Theory. Next, we collect useful results on Ollivier's discrete Ricci curvature for graphs and its modification by Lin, Lu, and Yau.

2.1 Terminology and notations

Throughout the paper, we use the following notations. Let $G = (V, E)$ be an undirected simple locally finite graph and V, E respectively denote the set of vertices and edges of G.

For $x \in V$ and $r \in \mathbb{N}$ we define the *r-sphere centered at x* as $S_r(x) = \{y \in V : d(x, y) = r\}$ and the *r-ball centered at x* as $B_r(x) = \{y \in V : d(x, y) \leq r\}$. Denote by $d_x = |S_1(x)|$ the *degree* of a vertex $x \in V$. The *minimum degree* of a graph G is denoted by $\delta(G) = \min_{x \in V} d_x$. We call a graph G *locally finite* if every vertex has finite degree. The graph is said to be *d-regular* if every vertex has the same degree d. Given an edge xy, we denote the set of common neighbors of x and y by $N_{xy} = S_1(x) \cap S_1(y)$.

The *distance* between two vertices x, y is denoted by $d(x, y)$. The *diameter* of G is denoted by $\mathrm{diam}(G) = \max_{x,y \in V} d(x, y)$. For two vertex sets X and Y, the distance between X and Y is denoted by

$$d(X, Y) := \min\{d(x, y) \mid x \in X, y \in Y\}.$$

When $X = \{x\}$, we simply write $d(x, Y)$ instead of $d(\{x\}, Y)$. For two disjoint vertex sets X and Y, we write $E(X, Y)$ for the set of edges with one end in X and the other in Y.

Given two graphs G and H, the *Cartesian product* (denoted by $G \square H$) is a graph on the vertex set $V(G) \times V(H)$, where two pairs (u_1, v_1) and (u_2, v_2) are adjacent if either $u_1 = u_2$ and $v_1 v_2 \in E(H)$, or $u_1 u_2 \in E(G)$ and $v_1 = v_2$.

2.2 Ollivier-Ricci and Lin–Lu–Yau curvature

We first recall the definition of Wasserstein distance, which is a metric on the space of probability measures and plays a fundamental role in optimal transport theory. For a graph $G = (V, E)$, a *probability measure* on G is a function $\mu : V \to [0, 1]$ such that $\sum_{x \in V} \mu(x) = 1$.

Definition 2.1 (Wasserstein distance)**.** Let $G = (V, E)$ be a locally finite graph. Let μ_1, μ_2 be two probability measures on G. The *Wasserstein distance* $W(\mu_1, \mu_2)$ between μ_1 and μ_2 is defined as

$$W(\mu_1, \mu_2) = \inf_{\pi} \sum_{x,y \in V} d(x, y)\pi(x, y),$$

where the infimum is taken over all the maps $\pi : V \times V \to [0, 1]$ satisfying

$$\mu_1(x) = \sum_{y \in V} \pi(x, y), \quad \forall x \in V$$

and

$$\mu_2(y) = \sum_{x \in V} \pi(x, y), \quad \forall y \in V.$$

Such a map is called a *transport plan* from μ_1 to μ_2. Imagine two distributions, μ_1 and μ_2, as piles of earth. A transport plan π transforms one pile of earth into another. The minimal effort which is required

for such a transformation is measured by the Wasserstein distance. If the infimum is attained, we call π an *optimal transport plan* transporting μ_1 to μ_2.

The Kantorovich duality tells the following identity about Wasserstein distance (see, e.g., [20, Theorem 1.14]):

$$W(\mu_1, \mu_2) = \sup_f \sum_{x \in V} f(x)(\mu_1(x) - \mu_2(x)),$$

where the supremum is taken over all 1-Lipschitz functions f, which means

$$|f(x) - f(y)| \leq d(x, y), \quad \forall x, y \in V.$$

In optimal transport theory, it is well known that no mass needs to be transported when it is already shared between the two probability measures.

Lemma 2.1. *Let $G = (V, E)$ be a locally finite graph. Let μ_1, μ_2 be two probability measures on V. Then there exists an optimal transport plan π transporting μ_1 to μ_2 satisfying*

$$\pi(x, x) = \min\{\mu_1(x), \mu_2(x)\}$$

for all $x \in V$.

To each vertex $x \in V$ and idleness parameter $\alpha \in [0, 1]$, we associate a probability measure μ_x^α defined by

$$\mu_x^\alpha(v) = \begin{cases} \alpha, & \text{if } v = x; \\ \frac{1-\alpha}{d_x}, & \text{if } xv \in E; \\ 0, & \text{otherwise.} \end{cases}$$

Then, the Ollivier-Ricci curvature is defined as follows.

Definition 2.2 (Ollivier-Ricci curvature)**.** Let $G = (V, E)$ be a locally finite graph. For any two distinct vertices x and y, the *α-Ollivier curvature* $\kappa_\alpha(x, y)$, $\alpha \in [0, 1]$, is defined as

$$\kappa_\alpha(x, y) = 1 - \frac{W(\mu_x^\alpha, \mu_y^\alpha)}{d(x, y)}.$$

The parameter α is called the *idleness*.

Following [11], we present a more intuitive formulation of the Ollivier-Ricci curvature. To this end, denote by ν_i^α the mass transported with distance i under an optimal transport plan transporting μ_x^α to μ_y^α. Then

$$\sum_{i=0}^{3} \nu_i^\alpha = 1 \quad \text{and} \quad W(\mu_x^\alpha, \mu_y^\alpha) = \sum_{i=1}^{3} i\nu_i^\alpha.$$

Hence, for any edge $xy \in E$, we conclude

$$\kappa_\alpha(x, y) = \nu_0^\alpha - \nu_2^\alpha - 2\nu_3^\alpha.$$

This formula plays a crucial role in proving our results.

Recall that the α-Ollivier-Ricci curvature depends on the idleness parameter. Ollivier initially considered the cases $\alpha = 0$ and $\alpha = \frac{1}{2}$, but its behavior across the full range of α remained an open question. Lin, Lu, and Yau [15] first established that κ_α is concave in $\alpha \in [0, 1]$. Using $\kappa_1(x, y) = 0$, one obtains that the function $h(\alpha) = \frac{\kappa_\alpha(x,y)}{1-\alpha}$ is increasing on the interval $[0, 1)$. By further proving that $h(\alpha)$ is bounded, they concluded that the limit $\lim_{\alpha \to 1} h(\alpha)$ exists. Building on this result, Lin, Lu, and Yau introduced a modified version of the Ollivier-Ricci curvature, obtained by differentiating κ_α with respect to the idleness parameter α.

Definition 2.3 (Lin–Lu–Yau curvature)**.** Let $G = (V, E)$ be a locally finite graph. For any two distinct vertices x and y, the *Lin–Lu–Yau curvature* is defined as

$$\kappa_{\mathrm{LLY}}(x, y) = \lim_{\alpha \to 1} \frac{\kappa_\alpha(x, y)}{1 - \alpha}.$$

In [2], Bourne et al. conducted a more detailed analysis of how κ_α depends on the idleness parameter α, by introducing the Ollivier-Ricci idleness function $\alpha \mapsto \kappa_\alpha(x, y)$. They demonstrated that the idleness function is piecewise linear with at most three linear parts. Additionally, they determined the length of the final linear part.

Theorem 2.1 ([2], Theorem 4.4)**.** *Let $G = (V, E)$ be a locally finite graph and let $x, y \in V$, $xy \in E$ with $d_x \geq d_y$. Then $\alpha \mapsto \kappa_\alpha(x, y)$ is linear over $\left[\frac{1}{d_x+1}, 1\right]$.*

As a direct consequence of the mean value theorem, they derived the following relation between the α-Ollivier-Ricci curvature and its modification by Lin, Lu, and Yau.

Theorem 2.2. *Let $G = (V, E)$ be a locally finite graph and let $x, y \in V$, $xy \in E$ with $d_x \geq d_y$. Then*

$$\kappa_\alpha(x, y) = (1 - \alpha)\kappa_{\mathrm{LLY}}(x, y)$$

for $\alpha \in \left[\frac{1}{d_x+1}, 1\right]$.

Loisel and Romon had previously established this relation in [16] for $\alpha \geq \frac{1}{2}$ in 2014. Consequently, for large values of α, the Lin–Lu–Yau curvature aligns with the α-Ollivier-Ricci curvature up to a scaling factor.

In the case of regular graphs, the optimal transport problem can be reduced to an optimal assignment problem between subsets of the spheres. This observation was leveraged in [13] to derive a more concise expression for the Lin–Lu–Yau curvature.

Theorem 2.3 ([13], Theorem 1.1)**.** *Let G be a locally finite graph. Let x and y be two adjacent vertices of equal degree d. Then the Lin–Lu–Yau curvature is as follows:*

$$\kappa_{\mathrm{LLY}}(x, y) = \frac{1}{d}\left(d + 1 - \min_{\phi: N_x \to N_y} \sum_{v \in N_x} d(v, \phi(v))\right),$$

where the minimum is taken over all the bijections ϕ from $N_x := S_1(x) \setminus B_1(y)$ to $N_y := S_1(y) \setminus B_1(x)$.

We define the lower Lin–Lu–Yau curvature bound of a graph at various scales as follows.

Definition 2.4. Let $G = (V, E)$ be a locally finite graph. For any positive integer i, we define the lower Lin–Lu–Yau curvature bound of G at scale i as follows:

$$\kappa^{(i)}_{\mathrm{LLY}}(G) := \inf_{\substack{x,y \in V \\ d(x,y)=i}} \kappa_{\mathrm{LLY}}(x, y).$$

When $i = 1$, we simply write $\kappa_{\mathrm{LLY}}(G)$ instead of $\kappa^{(1)}_{\mathrm{LLY}}(G)$. The following lemma is a reformulation of [15, Lemma 2.3], which is proved by the triangle inequality of Wasserstein distance.

Lemma 2.2. *Let G be a locally finite graph. For any positive integer i, it holds that*

$$\kappa^{(i)}_{\mathrm{LLY}}(G) \geq \kappa_{\mathrm{LLY}}(G).$$

Chen, Liu and You proved that positive Lin–Lu–Yau curvature implies large connectivity. They further characterized the structure of the infinite graph family for which equality is attained in the inequality.

Theorem 2.4 ([7], Theorem 1.1)**.** *Let G be a non-complete connected graph with minimum degree $\delta(G)$ and connectivity $\kappa(G)$. Then,*

$$\kappa(G) \geq \delta(G)\kappa^{(2)}_{\mathrm{LLY}}(G).$$

Hehl established a lower bound on the minimum degree that guarantees non-negative Lin–Lu–Yau curvature. He also proceed to discuss the sharpness of the bound.

Theorem 2.5 ([12], Theorem 1.1)**.** *Let G be a connected graph with n vertices and minimum degree $\delta(G)$. If*

$$\delta(G) \geq \frac{2n}{3} - 1,$$

then $\kappa_{\mathrm{LLY}}(x, y) \geq 0$ for all edges xy in G.

Theorem 2.6 ([12], Theorem 1.2)**.** *Let $n \in \mathbb{N}$ such that $2n/3 \in 2\mathbb{N}$ and $2n/3 \geq 6$. Then, there exists a graph G with n vertices and minimum degree*

$$\delta(G) = \frac{2n}{3} - 2,$$

that contains an edge xy with $\kappa_{\mathrm{LLY}}(x, y) < 0$.

3 Positive Curvature Implies Large Edge-Connectivity

In this section, we prove sharp lower bounds of edge-connectivity based on Lin–Lu–Yau curvature. Our results show that a large positive lower bound of Lin–Lu–Yau curvature on each edge (or between any two vertices at distance three) implies large edge-connectivity. Before our proof, we present some basic lemmas.

Lemma 3.1. *Let G be a non-complete connected graph with edge-connectivity $\kappa'(G)$. Let $S \subset E$ be an edge set of size $|S| = \kappa'(G)$ such that $G - S$ is disconnected and X, Y be two distinct connected components of $G - S$. Let $X_1 := X \cap V(S)$, $X_2 := X \setminus X_1$, $Y_1 := Y \cap V(S)$, $Y_2 := Y \setminus Y_1$. Then, for any edge $uw \in S$ with $u \in X_1$ and $w \in Y_1$, if $X_2 \neq \emptyset$, u has at least one neighbor in X_2; if $Y_2 \neq \emptyset$, w has at least one neighbor in Y_2.*

Proof. Without loss of generality, assume that $X_2 \neq \emptyset$. Suppose that $u \in X_1$ has no neighbors in X_2. Then $G - (S \setminus \{uw\})$ is disconnected, which is contradictory to the minimality of S. □

Lemma 3.2. *Let $G \cong K_\kappa \square K_2$ be the Cartesian product of K_κ and K_2. For any two vertices $x, y \in V(G)$ satisfying $d(x, y) = 2$, we have $\kappa_{\mathrm{LLY}}(x, y) = (\kappa + 2)/2\kappa$. In particular, $\kappa^{(2)}_{\mathrm{LLY}}(G) = (\kappa + 2)/2\kappa$.*

Proof. Write $V(G) = [\kappa] \times \{0, 1\}$. Without loss of generality, assume that $x = (1, 0)$, $y = (2, 1)$. Define $a = (1, 1)$, $b = (2, 0)$, and $u_i = (i, 0)$, $v_i = (i, 1)$ for $i = 3, \ldots, \kappa$. Then $S_1(x) = \{a, b, u_3, \ldots, u_\kappa\}$ and $S_1(y) = \{a, b, v_3, \ldots, v_\kappa\}$. Hence $S_1(x) \cap S_1(y) = \{a, b\}$ and u_i is adjacent to v_i for $i = 3, \ldots, \kappa$. Since G is κ-regular, setting $q := (1 - p)/\kappa$.

Consider the transport plan π from μ^p_x to μ^p_y given by

$$\begin{aligned} \pi(x, y) &= p, \\ \pi(a, a) &= \pi(b, b) = q, \\ \pi(u_i, v_i) &= q, \text{ for } i = 3, \ldots, \kappa. \end{aligned}$$

Its transport cost is

$$\mathrm{Cost}(\pi) = 2p + (\kappa - 2)q = 2p + \frac{\kappa - 2}{\kappa}(1 - p).$$

Therefore

$$W(\mu^p_x, \mu^p_y) \leq 2p + \frac{\kappa - 2}{\kappa}(1 - p).$$

For $v \in V(G)$, we have the following:

$$\begin{aligned} d(v, y) &= 2, \text{ for } v = x; \\ d(v, y) &= 2, \text{ for } v = u_i, i = 3, \ldots, \kappa; \\ d(v, y) &= 1, \text{ for } v = a, b \text{ and } v_i, i = 3, \ldots, \kappa. \end{aligned}$$

To prove optimality, define the 1-Lipschitz function $f(v) := d(v, y)$. According to Kantorovich duality, we derive

$$\begin{aligned} W(\mu^p_x, \mu^p_y) &= \sup_{\substack{f: V \to \mathbb{R} \\ 1-\text{Lipschitz}}} \sum_{v \in V} f(v) \left(\mu^p_x(v) - \mu^p_y(v) \right) \\ &\geq \sum_{v \in V} d(v, y) \mu^p_x(v) - \sum_{v \in V} d(v, y) \mu^p_y(v) \\ &= 2p + (2\kappa - 2)q - \kappa q \\ &= 2p + (\kappa - 2)q \\ &= 2p + \frac{\kappa - 2}{\kappa}(1 - p). \end{aligned}$$

Combining with the previous upper bound gives

$$W(\mu_x^p, \mu_y^p) = 2p + \frac{\kappa - 2}{\kappa}(1 - p).$$

By the definition of Lin–Lu–Yau curvature, we have

$$\kappa_{\mathrm{LLY}}(x, y) = \lim_{p \to 1} \frac{1}{1 - p} \left(1 - \frac{W(\mu_x^p, \mu_y^p)}{d(x, y)}\right) = \frac{\kappa + 2}{2\kappa}.$$

Since every pair of vertices at distance two in $K_\kappa \square K_2$ belongs to the same automorphism orbit, it follows that

$$\kappa_{\mathrm{LLY}}^{(2)}(K_\kappa \square K_2) = \frac{\kappa + 2}{2\kappa}.$$

We complete the proof of Lemma 3.2. □

Proof of Theorem 1.1. Let $G = (V, E)$ be a non-complete connected graph with edge-connectivity $\kappa'(G)$. Let $S \subset E$ be a edge set of size $|S| = \kappa'(G)$ such that $G - S$ is disconnected. Let X and Y be two distinct connected components of $G - S$. Let us set

$$X_1 := X \cap V(S),\ X_2 := X \setminus X_1,\ Y_1 := Y \cap V(S),\ Y_2 := Y \setminus Y_1.$$

Since $\mathrm{diam}(G) = 2$, at least one of X_2 and Y_2 is empty. We divide the discussion into two cases according to whether both X_2 and Y_2 are empty.

Case 1: One of X_2 and Y_2 is empty.

Without loss of generality, assume that $X_2 \neq \emptyset,\ Y_2 = \emptyset$. Let u be a vertex in X_1. Let x be a neighbor of u in X_2 (by Lemma 3.1, such x must exist). Let w be a neighbor of u in Y_1. Then, we have the following:

$$\begin{aligned} d(v, X_2) &= 1, \text{ for any } v \in S_1(x) \cap X_1; \\ d(v, X_2) &= 1, \text{ for any } v \in S_1(w) \cap X_1; \\ d(v, X_2) &= 2, \text{ for any } v \in S_1(w) \cap Y_1. \end{aligned}$$

Define the 1-Lipschitz function $f(v) := d(v, X_2)$. According to Kantorovich duality, we derive

$$\begin{aligned} W(\mu_w^p, \mu_x^p) &= \sup_{\substack{f: V \to \mathbb{R} \\ 1\text{–Lipschitz}}} \sum_{v \in V} f(v) \left(\mu_w^p(v) - \mu_x^p(v)\right) \\ &\geq \sum_{v \in V} d(v, X_2)\mu_w^p(v) - \sum_{v \in V} d(v, X_2)\mu_x^p(v) \\ &= \frac{|S_1(w) \cap X_1|(1 - p)}{d_w} + \left(p + \frac{|S_1(w) \cap Y_1|(1 - p)}{d_w}\right) \times 2 - \frac{|S_1(x) \cap X_1|(1 - p)}{d_x} \\ &= \frac{|S_1(w) \cap X_1|(1 - p)}{d_w} + \frac{2p|S_1(w) \cap X_1| + 2|S_1(w) \cap Y_1|}{d_w} - \frac{|S_1(x) \cap X_1|(1 - p)}{d_x} \\ &= \frac{2|S_1(w) \cap X_1| + 2|S_1(w) \cap Y_1|}{d_w} + \frac{|S_1(w) \cap X_1|(p - 1)}{d_w} - \frac{|S_1(x) \cap X_1|(1 - p)}{d_x} \\ &= 2 - \frac{|S_1(w) \cap X_1|(1 - p)}{d_w} - \frac{|S_1(x) \cap X_1|(1 - p)}{d_x} \\ &\geq 2 - \frac{1 - p}{\delta(G)} \left(|S_1(w) \cap X_1| + |S_1(x) \cap X_1|\right). \end{aligned}$$

In the above equality, we have employed the fact that $d_w = |S_1(w) \cap X_1| + |S_1(w) \cap Y_1|$.

By the definition of Lin–Lu–Yau curvature, we estimate

$$\kappa_{\mathrm{LLY}}(w, x) = \lim_{p \to 1} \frac{1}{1 - p} \left(1 - \frac{W(\mu_w^p, \mu_x^p)}{d(x, w)}\right) \leq \frac{|S_1(w) \cap X_1| + |S_1(x) \cap X_1|}{2\delta(G)}.$$

Therefore, we derive

$$\begin{aligned} \kappa'(G) = |S| &\geq \max\{|S_1(x) \cap X_1|, |S_1(w) \cap X_1|\} \\ &\geq \frac{1}{2}(|S_1(x) \cap X_1| + |S_1(w) \cap X_1|) \geq \delta(G)\kappa_{\mathrm{LLY}}^{(2)}(G). \end{aligned}$$

We complete the proof of Case 1.

Case 2: Both X_2 and Y_2 are empty.

Let $\kappa := \kappa'(G)$. We first prove the following claim determining the structure of G.

Claim 1. If $X_2 = Y_2 = \emptyset$, then $G \cong K_\kappa \square K_2$.

Proof. We claim that the subgraphs induced by the vertex sets X_1 and Y_1 are both complete graphs. Suppose not. Without loss of generality, choose $u \in X_1$. Since $d_u \geq \delta(G) \geq \kappa$ and the number of neighbors of u lying in X_1 is at most $\kappa - 2$ because $|X_1| \leq \kappa$, it follows that $|E(u, Y_1)| \geq 2$, and therefore

$$|E(X_1, Y_1)| = |E(u, Y_1)| + |E(X_1 \setminus \{u\}, Y_1)| \geq 2 + (\kappa - 1) = \kappa + 1,$$

contradicting the definition of $\kappa'(G)$.

Now we claim that $|X_1| = |Y_1| = \kappa$. Note that $|X_1| \leq \kappa$ by the definition of $\kappa'(G)$. For any vertex $u \in X_1$,

$$d_u = |X_1| - 1 + |E(u, Y_1)| \leq |X_1| - 1 + (\kappa - |X_1| + 1) = \kappa.$$

Since $d_u \geq \delta(G) \geq \kappa$, we have $d_u = \kappa$ and $|E(u, Y_1)| = \kappa - |X_1| + 1$. This holds for every $u \in X_1$. It follows that

$$\kappa = |E(X_1, Y_1)| = \sum_{u \in X_1} |E(u, Y_1)| = |X_1|(\kappa - |X_1| + 1).$$

Solving the equation yields $|X_1| = \kappa$ or $|X_1| = 1$, where the latter is excluded by our assumptions. By symmetry, a similar argument applied to vertices in Y_1 yields $|Y_1| = \kappa$ as well.

Since the subgraphs induced by X_1 and Y_1 are both complete graphs and $|X_1| = |Y_1| = \kappa$, the edges between X_1 and Y_1 must form a perfect matching. Consequently, $G \cong K_\kappa \square K_2$. We complete the proof of Claim 1. □

Let $x, y \in V(K_\kappa \square K_2)$ and $d(x, y) = 2$, we have $\kappa^{(2)}_{\mathrm{LLY}}(K_\kappa \square K_2) = (\kappa+2)/2\kappa$ by Lemma 3.2 and $\delta(K_\kappa \square K_2) = \kappa \geq 2$. Therefore, we derive

$$\begin{aligned} \kappa'(G) &= 2\delta(G)\kappa^{(2)}_{\mathrm{LLY}}(G) - 2 \\ &= \delta(G)\kappa^{(2)}_{\mathrm{LLY}}(G) + \frac{\kappa - 2}{2} \geq \delta(G)\kappa^{(2)}_{\mathrm{LLY}}(G). \end{aligned}$$

We complete the proof of Theorem 1.1. □

Proof of Theorem 1.2. Let $G = (V, E)$ be a non-complete connected graph with edge-connectivity $\kappa'(G)$. Let $S \subset E$ be a edge set of size $|S| = \kappa'(G)$ such that $G - S$ is disconnected. Let X and Y be two distinct connected components of $G - S$. Let us set

$$X_1 := X \cap V(S),\ X_2 := X \setminus X_1,\ Y_1 := Y \cap V(S),\ Y_2 := Y \setminus Y_1.$$

Since $\mathrm{diam}(G) \geq 3$, at most one of X_2 and Y_2 is empty. We divide the discussion into two cases according to whether both X_2 and Y_2 are not empty.

Case 1: One of X_2 and Y_2 is empty.

Without loss of generality, assume that $X_2 \neq \emptyset,\ Y_2 = \emptyset$. Let u be a vertex in X_1. Let x be a neighbor of u in X_2 (by Lemma 3.1, such x must exist). Let w be a neighbor of u in Y_1. Then, we have the following:

$$\begin{aligned} &d(v, X_2) = 1, \text{ for any } v \in S_1(x) \cap X_1; \\ &d(v, X_2) = 1, \text{ for any } v \in S_1(w) \cap X_1; \\ &d(v, X_2) = 2, \text{ for any } v \in S_1(w) \cap Y_1. \end{aligned}$$

Define the 1-Lipschitz function $f(v) := d(v, X_2)$. According to Kantorovich duality, we derive

$$\begin{aligned}
W(\mu_w^p, \mu_x^p) &= \sup_{\substack{f:V\to\mathbb{R}\\ 1-\text{Lipschitz}}} \sum_{v\in V} f(v)\left(\mu_w^p(v) - \mu_x^p(v)\right) \\
&\geq \sum_{v\in V} d(v, X_2)\mu_w^p(v) - \sum_{v\in V} d(v, X_2)\mu_x^p(v) \\
&= \frac{|S_1(w)\cap X_1|(1-p)}{d_w} + \left(p + \frac{|S_1(w)\cap Y_1|(1-p)}{d_w}\right)\times 2 - \frac{|S_1(x)\cap X_1|(1-p)}{d_x} \\
&= 2 - \frac{|S_1(w)\cap X_1|(1-p)}{d_w} - \frac{|S_1(x)\cap X_1|(1-p)}{d_x} \\
&\geq 2 - \frac{1-p}{\delta(G)}\left(|S_1(w)\cap X_1| + |S_1(x)\cap X_1|\right).
\end{aligned}$$

In the above equality, we have employed the fact that $d_w = |S_1(w)\cap X_1| + |S_1(w)\cap Y_1|$.

By the definition of Lin–Lu–Yau curvature, we estimate

$$\kappa_{\mathrm{LLY}}(w, x) = \lim_{p\to 1}\frac{1}{1-p}\left(1 - \frac{W(\mu_w^p, \mu_x^p)}{d(x, w)}\right) \leq \frac{|S_1(w)\cap X_1| + |S_1(x)\cap X_1|}{2\delta(G)}.$$

Therefore, we derive

$$\begin{aligned}
\kappa'(G) = |S| &\geq \max\{|S_1(x)\cap X_1|, |S_1(w)\cap X_1|\} \\
&\geq \frac{1}{2}(|S_1(x)\cap X_1| + |S_1(w)\cap X_1|) \geq \delta(G)\kappa_{\mathrm{LLY}}^{(2)}(G).
\end{aligned}$$

We complete the proof of Case 1.

Case 2: Both X_2 and Y_2 are not empty.

Let $uw \in S$ with $u \in X_1$ and $w \in Y_1$, let x be a neighbor of u in X_2 and y be a neighbor of w in Y_2 (by Lemma 3.1, such x and y must exist). Then, we have the following:

$$\begin{aligned}
&d(v, X_2) = 1, \text{ for any } v \in S_1(x)\cap X_1; \\
&d(v, X_2) = 2, \text{ for any } v \in S_1(y)\cap Y_1; \\
&d(v, X_2) \geq 3, \text{ for any } v \in S_1(y)\cap Y_2.
\end{aligned}$$

Define the 1-Lipschitz function $f(v) := d(v, X_2)$. According to Kantorovich duality, we derive

$$\begin{aligned}
W(\mu_y^p, \mu_x^p) &= \sup_{\substack{f:V\to\mathbb{R}\\ 1-\text{Lipschitz}}} \sum_{v\in V} f(v)\left(\mu_y^p(v) - \mu_x^p(v)\right) \\
&\geq \sum_{v\in V} d(v, X_2)\mu_y^p(v) - \sum_{v\in V} d(v, X_2)\mu_x^p(v) \\
&\geq \frac{|S_1(y)\cap Y_1|(1-p)}{d_y}\times 2 + \left(p + \frac{|S_1(y)\cap Y_2|(1-p)}{d_y}\right)\times 3 - \frac{|S_1(x)\cap X_1|(1-p)}{d_x} \\
&= \frac{|S_1(y)\cap Y_1|(2-2p)}{d_y} + \frac{3p|S_1(y)\cap Y_1| + 3|S_1(y)\cap Y_2|}{d_y} - \frac{|S_1(x)\cap X_1|(1-p)}{d_x} \\
&= \frac{3|S_1(y)\cap Y_1| + 3|S_1(y)\cap Y_2|}{d_y} + \frac{|S_1(y)\cap Y_1|(p-1)}{d_y} - \frac{|S_1(x)\cap X_1|(1-p)}{d_x} \\
&= 3 - \frac{|S_1(y)\cap Y_1|(1-p)}{d_y} - \frac{|S_1(x)\cap X_1|(1-p)}{d_x} \\
&\geq 3 - \frac{1-p}{\delta(G)}\left(|S_1(y)\cap Y_1| + |S_1(x)\cap X_1|\right).
\end{aligned}$$

In the above equality, we have employed the fact that $d_y = |S_1(y)\cap Y_1| + |S_1(y)\cap Y_2|$.

By the definition of Lin–Lu–Yau curvature, we estimate

$$\kappa_{\mathrm{LLY}}(y, x) = \lim_{p\to 1}\frac{1}{1-p}\left(1 - \frac{W(\mu_y^p, \mu_x^p)}{d(x, y)}\right) \leq \frac{|S_1(y)\cap Y_1| + |S_1(x)\cap X_1|}{3\delta(G)}.$$

Therefore, we derive

$$\begin{aligned}\kappa'(G) = |S| &\geq \max\{|S_1(x) \cap X_1|, |S_1(y) \cap Y_1|\} \\ &\geq \frac{1}{2}(|S_1(x) \cap X_1| + |S_1(y) \cap Y_1|) = \frac{3}{2} \times \frac{|S_1(x) \cap X_1| + |S_1(y) \cap Y_1|}{3} \\ &\geq \frac{3}{2}\delta(G)\kappa^{(3)}_{\mathrm{LLY}}(G).\end{aligned}$$

We complete the proof of Theorem 1.2. □

The next lemma characterized the structure of the infinite family of graphs for which Theorem 1.2 is sharp.

Lemma 3.3. *Let $G_{2n} \cong K_n \cup K_n + M_2$ be the union of two copies of the complete graph K_n ($n \geq 3$) with a matching of size 2 between them. For any two vertices $x, y \in V(G_{2n})$ satisfying $d(x, y) = 3$, we have $\kappa_{\mathrm{LLY}}(x, y) = 4/3(n-1)$. In particular, $\kappa^{(3)}_{\mathrm{LLY}}(G_{2n}) = 4/3(n-1)$.*

Proof. Without loss of generality, write

$$V(G_{2n}) = A \cup B = \{x, a_1, a_2, u_1, \ldots, u_{n-3}\} \cup \{y, b_1, b_2, v_1, \ldots, v_{n-3}\},$$

where the two edges of the matching are a_1b_1 and a_2b_2. Note that $d(x, y) = 3$, $d(a_i, b_i) = 1$ for $i = 1, 2$ and $d(u_i, v_j) = 3$ for $i = 1, \ldots, n-3$. Since $d_x = d_y = n-1$, setting $q := (1-p)/(n-1)$.

Consider the transport plan π from μ^p_x to μ^p_y given by

$$\begin{aligned}\pi(x, y) &= p, \\ \pi(a_1, b_1) = \pi(a_2, b_2) &= q, \\ \pi(u_i, v_i) &= q, \text{ for } i = 1, \ldots, n-3.\end{aligned}$$

Its transport cost is

$$\mathrm{Cost}(\pi) = 3p + 2q + 3(n-3)q = 3 - \frac{4}{n-1}(1-p).$$

Therefore

$$W(\mu^p_x, \mu^p_y) \leq 3 - \frac{4}{n-1}(1-p).$$

To prove optimality, define the 1-Lipschitz function $f : V(G_{2n}) \to \mathbb{R}$ by

$$f(z) = \begin{cases} 2, & z \in A \setminus \{a_1, a_2\}; \\ 1, & z \in \{a_1, a_2\}; \\ 0, & z \in \{b_1, b_2\}; \\ -1, & z \in B \setminus \{b_1, b_2\}. \end{cases}$$

According to Kantorovich duality, we derive

$$\begin{aligned}W(\mu^p_x, \mu^p_y) &= \sup_{\substack{f: V(G_{2n}) \to \mathbb{R} \\ 1\text{–Lipschitz}}} \sum_{z \in V(G_{2n})} f(z)\left(\mu^p_x(z) - \mu^p_y(z)\right) \\ &\geq \sum_{z \in V(G_{2n})} f(z)\mu^p_x(z) - \sum_{z \in V(G_{2n})} f(z)\mu^p_y(z) \\ &= (2p + 2q + 2(n-3)q) - (-p - (n-3)q) \\ &= 3p + (3n-7)q \\ &= 3 - \frac{4}{n-1}(1-p).\end{aligned}$$

Combining with the previous upper bound gives

$$W(\mu^p_x, \mu^p_y) = 3 - \frac{4}{n-1}(1-p).$$

By the definition of Lin–Lu–Yau curvature, we have

$$\kappa_{\mathrm{LLY}}(x,y) = \lim_{p\to 1} \frac{1}{1-p}\left(1 - \frac{W(\mu_x^p, \mu_y^p)}{d(x,y)}\right) = \frac{4}{3(n-1)}.$$

Since every pair of vertices at distance three in G_{2n} belongs to the same automorphism orbit, it follows that

$$\kappa_{\mathrm{LLY}}^{(3)}(G_{2n}) = \frac{4}{3(n-1)}.$$

We complete the proof of Lemma 3.3. □

4 Large Edge-Connectivity Implies Large Curvature

In this section, we give two sharp lower bounds of Lin–Lu–Yau curvature on each edge. These results show that large edge-connectivity can guarantee large Lin–Lu–Yau curvature on each edge, which complements the conclusion of Section 3. Before presenting our proof, we require the following lemma, which guarantees that the diameter of a graph is at most two when the minimum degree is sufficiently large.

Lemma 4.1. *Let G be a graph on n vertices. If the minimum degree $\delta(G)$ of G satisfies*

$$\delta(G) \geq \frac{n-1}{2},$$

then the diameter of G satisfies $\mathrm{diam}(G) \leq 2$.

Proof. Assume that $\mathrm{diam}(G) > 2$. Hence, there exist vertices $x, y \in V$ such that $d(x,y) > 2$, and therefore $S_1(x) \cap S_1(y) = \emptyset$. As $x, y \notin S_1(x) \cup S_1(y)$, we have

$$|S_1(x) \cup S_1(y)| \leq n-2.$$

Nevertheless, we have

$$|S_1(x) \cup S_1(y)| = d_x + d_y \geq 2\delta(G) \geq n-1.$$

Therefore, our assumption was incorrect, and it follows that $\mathrm{diam}(G) \leq 2$ must hold. □

We also require the following lemma, which determines $\kappa'(G)$ when the minimum degree is sufficiently large.

Lemma 4.2. *If a graph satisfies $\delta(G) \geq (n-1)/2$, then $\kappa'(G) = \delta(G)$.*

Proof. Let S be a non-empty proper vertex subset, and let $s := |S| \leq n/2$. Every vertex of S has at most $s-1$ neighbors inside S, hence at least $(\delta(G) - s + 1)$ neighbors outside S. Therefore,

$$E(S, V-S) \geq s(\delta(G) - s + 1).$$

Since $s \leq \delta(G)$,

$$s(\delta(G) - s + 1) \geq \delta(G).$$

Thus every edge-cut has size at least $\delta(G)$. On the other hand, removing all edges incident with a minimum-degree vertex disconnects the graph and uses exactly $\delta(G)$ edges. Hence

$$\kappa'(G) = \delta(G).$$

We complete the proof of Lemma 4.2. □

Proof of Theorem 1.3. Let xy be an arbitrary edge in G and assume, without loss of generality, that $d_x \leq d_y$. Choose

$$\alpha = \frac{1}{d_x + 1}.$$

Let π be an optimal transport plan, transporting μ_x^α to μ_y^α. Denote by ν_i^α the mass transported with distance i under π. According to Lemma 4.1 and the fact that $\delta(G) \geq \kappa'(G)$, we have $\nu_3^\alpha = 0$ and therefore

$$\nu_0^\alpha + \nu_1^\alpha + \nu_2^\alpha = 1.$$

Thus, we obtain

$$\begin{aligned}\kappa_\alpha(x,y) &= 1 - W_1(\mu_x^\alpha, \mu_y^\alpha) \\ &= \nu_0^\alpha + \nu_1^\alpha + \nu_2^\alpha - \nu_1^\alpha - 2\nu_2^\alpha \\ &= \nu_0^\alpha - \nu_2^\alpha.\end{aligned}$$

According to Lemma 2.1, we may assume

$$\pi(z,z) = \min\{\mu_x^\alpha(z), \mu_y^\alpha(z)\}$$

for every $z \in V$. Therefore, $\pi(z,z) = (1-\alpha)/d_y$ for every $z \in N_{xy}$, $\pi(x,x) = (1-\alpha)/d_y$, and $\pi(y,y) = (1-\alpha)/d_x$. This yields

$$\nu_0^\alpha = \frac{1-\alpha}{d_y}(|N_{xy}| + 1) + \frac{1-\alpha}{d_x}.$$

Since $\nu_2^\alpha \leq 1 - \nu_0^\alpha$, we conclude

$$\begin{aligned}\kappa_\alpha(x,y) &\geq 2\nu_0^\alpha - 1 \\ &= 2(1-\alpha)\left[\frac{|N_{xy}|+1}{d_y} + \frac{1}{d_x} - \frac{1}{2(1-\alpha)}\right] \\ &= 2(1-\alpha)\left[\frac{|N_{xy}|+1}{d_y} + \frac{1}{d_x} - \frac{d_x+1}{2d_x}\right] \\ &= 2(1-\alpha)\left[\frac{|N_{xy}|+1}{d_y} + \frac{1}{2d_x} - \frac{1}{2}\right] \\ &\geq 2(1-\alpha)\left[\frac{2|N_{xy}|+2}{2d_y} + \frac{1}{2d_y} - \frac{1}{2}\right] \\ &= (1-\alpha)\left[\frac{2|N_{xy}|+3}{d_y} - 1\right].\end{aligned}$$

According to Theorem 2.2, the equality $\kappa_\alpha(x,y) = (1-\alpha)\kappa_{\mathrm{LLY}}(x,y)$ holds. Hence,

$$\kappa_{\mathrm{LLY}}(x,y) \geq \frac{2|N_{xy}| + 3 - d_y}{d_y}.$$

We next prove that the right-hand side of the inequality is strictly positive.

Since $d_x, d_y \geq \delta(G)$, we have

$$|N_{xy}| \geq d_x + d_y - n \geq 2\delta(G) - n \geq 2\kappa'(G) - n.$$

Furthermore,

$$d_y \leq n - d_x + |N_{xy}| \leq n - \kappa'(G) + |N_{xy}|.$$

Putting the inequalities together yields

$$\begin{aligned}2|N_{xy}| + 3 - d_y &\geq |N_{xy}| + \kappa'(G) + 3 - n \\ &\geq (2\kappa'(G) - n) + \kappa'(G) + 3 - n \\ &= 3\kappa'(G) + 3 - 2n.\end{aligned}$$

When $\kappa'(G) > 2n/3 - 1$, then $2|N_{xy}| + 3 - d_y \geq 3\kappa'(G) + 3 - 2n > 0$, yielding $\kappa_{\mathrm{LLY}}(x,y) > 0$.

When $\kappa'(G) = 2n/3 - 1$, since $\kappa'(G)$ is an integer, we may assume that

$$n = 3r, \qquad \kappa'(G) = 2r - 1.$$

Suppose that equality $2|N_{xy}|+3-d_y = 3\kappa'(G)+3-2n$ holds, so that $2|N_{xy}|+3-d_y = 0$ (If not, $2|N_{xy}|+3-d_y > 0$ yields $\kappa_{\mathrm{LLY}}(x,y) > 0$). Then every inequality above is forced to be equality, we have $|N_{xy}| = 2\kappa'(G) - n$ and $d_y = n - \kappa'(G) + |N_{xy}|$. Hence,

$$d_x = d_y = \kappa'(G) = 2r - 1, \qquad |N_{xy}| = r - 2.$$

Define $N_x := S_1(x) \setminus B_1(y)$ and $N_y := S_1(y) \setminus B_1(x)$, then $|N_x| = |N_y| = r$. We claim that there must exist an edge between N_x and N_y. Indeed, suppose not. Choose $u \in N_x$, since $\mathrm{diam}(G) \leq 2$, then all neighbors of u lie in $\{x\} \cup (N_x \setminus \{u\}) \cup N_{xy}$, and therefore

$$d_u \leq 1 + (r-1) + (r-2) = 2r - 2 < 2r - 1 = \delta(G),$$

contradicting the definition of $\delta(G)$. Hence there exists $uv \in E(G)$ with $u \in N_x$ and $v \in N_y$. Therefore

$$\min_{\phi: N_x \to N_y} \sum_{v \in N_x} d(v, \phi(v)) \leq 1 + 2(r-1) = 2r - 1.$$

Since $d_x = d_y = 2r - 1$, according to Theorem 2.3,

$$\kappa_{\mathrm{LLY}}(x,y) \geq \frac{1}{2r-1}\Big((2r-1) + 1 - (2r-1)\Big) = \frac{1}{2r-1} > 0.$$

Since the edge xy was chosen arbitrarily, every edge has positive curvature. We complete the proof of Theorem 1.3. □

Next, we discuss the sharpness of the lower bound.

Proof of Theorem 1.4. Let $l \geq 2$ and define the vertex set

$$V = \{x, y, v\} \sqcup Z \sqcup X \sqcup Y = \{x, y, v, z_1, \ldots, z_{l-2}, x_1, \ldots, x_{l+1}, y_1, \ldots, y_{l+1}\},$$

where $|Z| = l - 2$, $|X| = |Y| = l + 1$. The graph is defined as follows:

- $xy \in E$;
- x is adjacent to all vertices of $Z \cup X$;
- y is adjacent to all vertices of $Z \cup Y$;
- every vertex of Z is adjacent to all vertices of $X \cup Y$;
- X induces a clique, Y induces a clique;
- v is adjacent to all vertices of $X \cup Y$.

Note that there are no vertices z_i in the case where $l = 2$. The vertices x and y are of degree $2l$ and the vertex v is of degree $2l + 2$. Additionally, we have

$$\begin{aligned} d_{x_i} &= 1 + (l-2) + l + 1 = 2l, \quad \forall i \in \{1, \ldots, l+1\}; \\ d_{y_i} &= 1 + (l-2) + l + 1 = 2l, \quad \forall i \in \{1, \ldots, l+1\}; \\ d_{z_i} &= 2 + 2(l+1) = 2l + 4, \quad \forall i \in \{1, \ldots, l-2\}. \end{aligned}$$

Therefore, the resulting graph satisfies $\delta(G) = 2l$, since $n = 3l + 3$ we have

$$\delta(G) = \frac{2n}{3} - 2.$$

Observe that

$$N_x := S_1(x) \setminus B_1(y) = \{x_1, \ldots, x_{l+1}\}, \; N_y := S_1(y) \setminus B_1(x) = \{y_1, \ldots, y_{l+1}\}.$$

Since $d_x = d_y$, we can apply Theorem 2.3. By construction, the equality $d(x_i, y_j) = 2$ holds for every $i, j \in \{1, \dots, l+1\}$. Hence, $d(z, \phi(z)) = 2$ for every $z \in S_1(x) \setminus B_1(y)$ and every bijection $\phi : S_1(x) \setminus B_1(y) \to S_1(y) \setminus B_1(x)$. Thus, we conclude

$$\begin{aligned}\kappa_{\mathrm{LLY}}(x,y) &= \frac{1}{d_x}\left(d_x + 1 - \inf_{\phi} \sum_{z \in S_1(x)\setminus B_1(y)} d(z, \phi(z))\right) \\ &= \frac{1}{d_x}\left(d_x + 1 - 2|S_1(x) \setminus B_1(y)|\right) \\ &= \frac{1}{2l}\left(2l + 1 - 2\,(l+1)\right) \\ &= -\frac{1}{2l}.\end{aligned}$$

Note that $\delta(G) = 2l \geq (3l+2)/2 = (n-1)/2$. Applying Lemma 4.2 to the above construction gives $\kappa'(G) = \delta(G) = 2l = 2n/3 - 2$. Therefore, there exist infinitely many graphs with $\kappa'(G) = 2n/3 - 2$ that contain an edge xy with $\kappa_{\mathrm{LLY}}(x, y) < 0$. We complete the proof of Theorem 1.4. □

Combining Lemma 4.2 with Theorem 1.3 yields the following corollary, which can be considered a strong extension of Theorem 2.5.

Corollary 4.1. *Let G be a connected graph with n vertices and minimum degree $\delta(G)$. If*

$$\delta(G) \geq \frac{2n}{3} - 1,$$

then $\kappa_{\mathrm{LLY}}(x, y) > 0$ for all edges xy in G.

We also derive a sufficient condition, depending only on the number of edges, for positive Lin–Lu–Yau curvature.

Proof of Theorem 1.5. Let $\kappa := \kappa'(G)$, since $\kappa \leq \delta(G)$, every vertex has degree at least κ. Hence $2m = \sum_{v \in V(G)} d_v \geq n\kappa$, therefore $n \leq 2m/\kappa$. By the condition $\kappa \geq 2n/3 - 1$ of Theorem 1.3, it suffices to prove that

$$\kappa \geq \frac{4m}{3\kappa} - 1.$$

After rearranging, $\kappa^2 + \kappa - 4m/3 \geq 0$. The positive root of $x^2 + x - 4m/3 = 0$ is

$$x = \frac{\sqrt{1 + \frac{16}{3}m} - 1}{2}.$$

Hence condition

$$\kappa \geq \left\lceil \frac{\sqrt{1 + \frac{16}{3}m} - 1}{2} \right\rceil$$

implies $\kappa^2 + \kappa - 4m/3 \geq 0$, and therefore $\kappa \geq 2n/3 - 1$. Applying Theorem1.3 yields $\kappa_{\mathrm{LLY}}(x, y) > 0$ for all edges xy in G. We complete the proof of Theorem 1.5. □

Remark 1. Asymptotically, the condition of Theorem 1.5 can be written as

$$\kappa'(G) \geq \frac{2}{\sqrt{3}}\sqrt{m} + O(1).$$

Thus an edge-connectivity of order $\sqrt{m}$ already forces positive Lin–Lu–Yau curvature.

Next, we discuss the sharpness of the lower bound.

Proof of Theorem 1.6. Let $t \geq 0$ and define the vertex set

$$V = \{x, y, v\} \sqcup A \sqcup X \sqcup Y = \{x, y, v, a_1, \ldots, a_t, x_1, \ldots, x_{t+2}, y_1, \ldots, y_{t+2}\},$$

where $|A| = t$, $|X| = |Y| = t + 2$. The graph is defined as follows:

- $xy \in E$;
- x is adjacent to all vertices of $A \cup X$;
- y is adjacent to all vertices of $A \cup Y$;
- every vertex of A is adjacent to all vertices of $X \cup Y$;
- X induces a clique, Y induces a clique;
- v is adjacent to all vertices of $X \cup Y$.

Note that there are no vertices a_i in the case where $t = 0$. The vertices x and y are of degree $2t + 3$ and the vertex v is of degree $2t + 4$. Additionally, we have

$$\begin{aligned} d_{x_i} &= 1 + t + (t + 1) + 1 = 2t + 3, \quad \forall i \in \{1, \ldots, t + 2\}; \\ d_{y_i} &= 1 + t + (t + 1) + 1 = 2t + 3, \quad \forall i \in \{1, \ldots, t + 2\}; \\ d_{a_i} &= 2 + 2(t + 2) = 2t + 6, \quad \forall i \in \{1, \ldots, t\}. \end{aligned}$$

Therefore, the resulting graph satisfies $\delta(G) = 2t + 3$, since $n = 3t + 7$ we have

$$\delta(G) = 2t + 3 \geq \frac{3t + 6}{2} = \frac{n - 1}{2}.$$

By Lemma 4.2, we have

$$\kappa'(G) = \delta(G) = 2t + 3.$$

Observe that

$$N_x := S_1(x) \setminus B_1(y) = \{x_1, \ldots, x_{t+2}\},\ N_y := S_1(y) \setminus B_1(x) = \{y_1, \ldots, y_{t+2}\}.$$

Since $d_x = d_y$, we can apply Theorem 2.3. By construction, the equality $d(x_i, y_j) = 2$ holds for every $i, j \in \{1, \ldots, l + 1\}$. Hence, $d(z, \phi(z)) = 2$ for every $z \in S_1(x) \setminus B_1(y)$ and every bijection $\phi : S_1(x) \setminus B_1(y) \to S_1(y) \setminus B_1(x)$. Thus, we conclude

$$\begin{aligned} \kappa_{\mathrm{LLY}}(x, y) &= \frac{1}{d_x}\left(d_x + 1 - \inf_{\phi} \sum_{z \in S_1(x) \setminus B_1(y)} d(z, \phi(z))\right) \\ &= \frac{1}{d_x}\left(d_x + 1 - 2|S_1(x) \setminus B_1(y)|\right) \\ &= \frac{1}{2t + 3}\left(2t + 3 + 1 - 2(t + 2)\right) \\ &= 0. \end{aligned}$$

As for the number of edges, we have

$$\begin{aligned} m &= 1 + 2(2t + 2) + 2t(t + 2) + 2\binom{t + 2}{2} + 2(t + 2) \\ &= 1 + 4t + 4 + 2t^2 + 4t + t^2 + 3t + 2 + 2t + 4 \\ &= 3t^2 + 13t + 11. \end{aligned}$$

Thus, we conclude

$$\kappa'(G) = 2t + 3 = (2t + 4) - 1 = \left\lfloor \frac{\sqrt{1 + \frac{16}{3}m} - 1}{2} \right\rfloor - 1.$$

Therefore there exist infinitely many graphs with $\kappa'(G) = \left\lfloor \frac{\sqrt{1+\frac{16}{3}m}-1}{2} \right\rfloor - 1$ that contain an edge xy with $\kappa_{\mathrm{LLY}}(x, y) = 0$. We complete the proof of Theorem 1.6. □

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